

```

> restart;
> with(plots) :
> f := x ->  $\frac{1}{\sigma \cdot \sqrt{2 \cdot \text{Pi}}} \cdot \exp\left(-\frac{(x - \mu x)^2}{2 \cdot \sigma^2}\right);$ 

```

$$f := x \rightarrow \frac{e^{-\frac{1}{2} \frac{(x - \mu x)^2}{\sigma^2}}}{\sigma \sqrt{2 \pi}}$$

(1)

```

> #standard.
>  $\mu x := 0; \sigma := 1;$ 

```

$$\mu x := 0$$

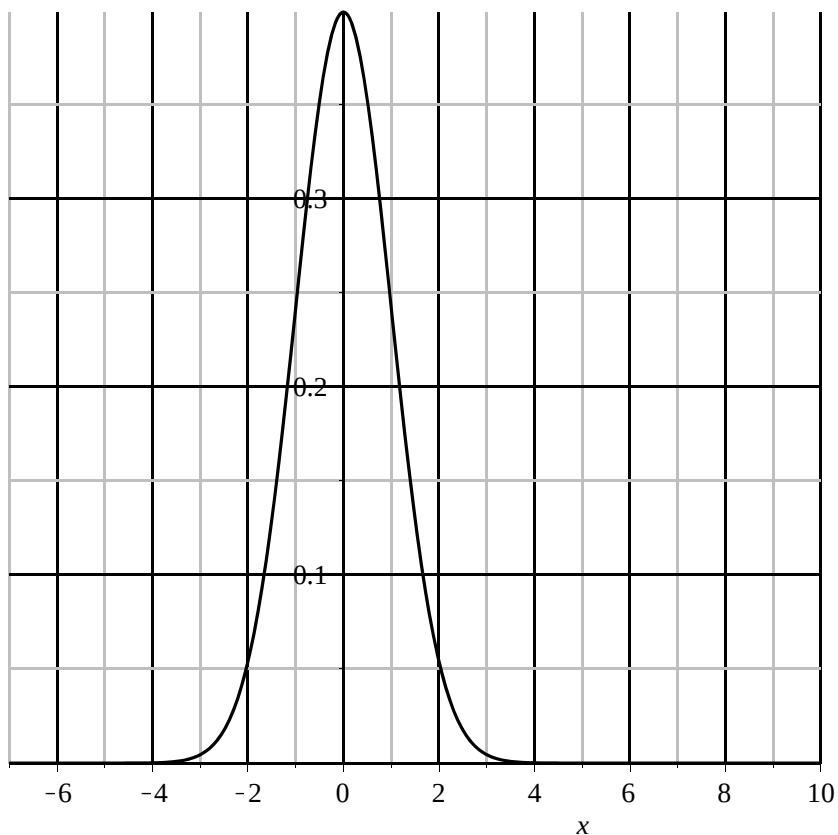
$$\sigma := 1$$

(2)

```

> q0 := plot( f(x), x=-7..10, color = black) :
> display(q0);

```



```

>  $\mu x := 5; \sigma := 1;$ 

```

$$\mu x := 5$$

$\sigma := 1$

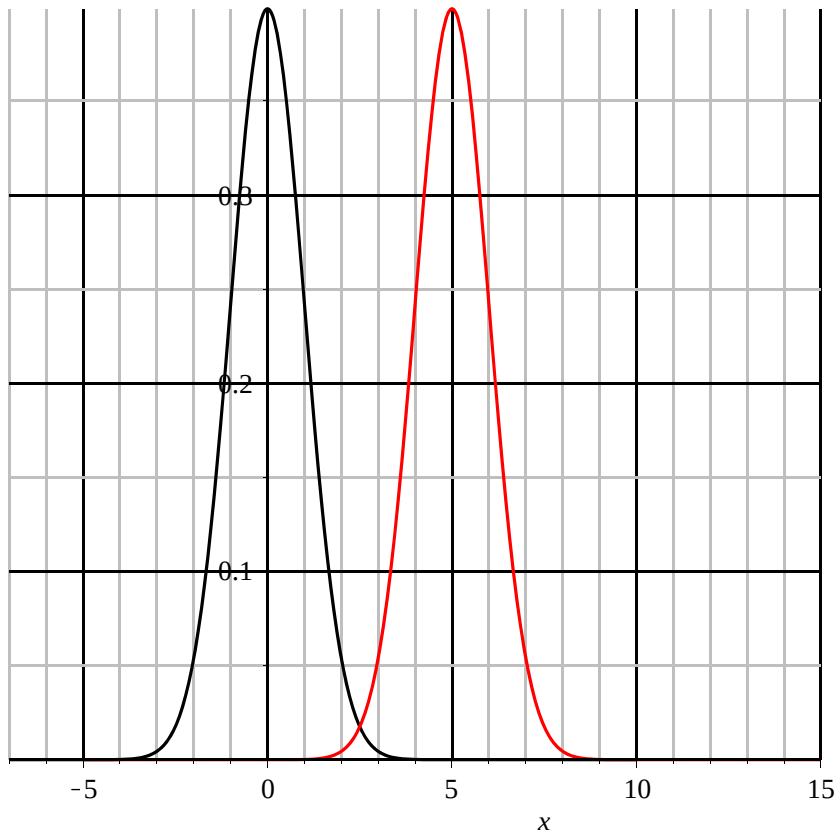
(3)

> $q1 := \text{plot}(f(x), x = -5..15, \text{color} = \text{red});$

$q1 := \text{PLOT}(\dots)$

(4)

> $\text{display}(q0, q1);$



> $\mu x := 0; \sigma := 3;$

$\mu x := 0$

$\sigma := 3$

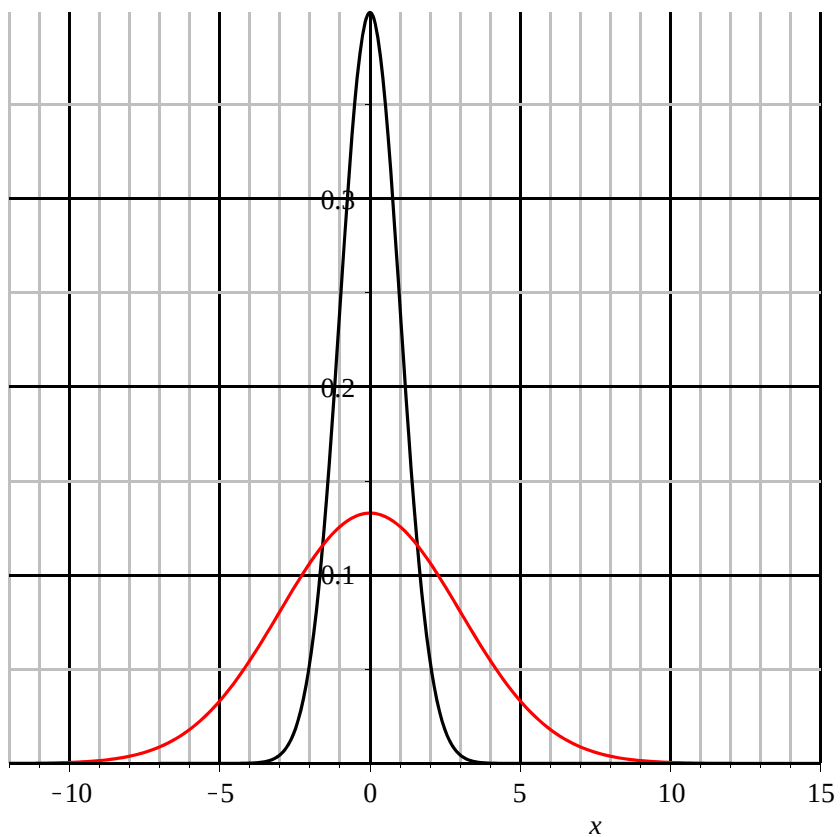
(5)

> $q1 := \text{plot}(f(x), x = -12..15, \text{color} = \text{red});$

$q1 := \text{PLOT}(\dots)$

(6)

> $\text{display}(q0, q1);$



```
> a := 2; σ := 1;
```

```
a := 2
```

```
σ := 1
```

(7)

```
> # Example 2.8
```

```
> μx := 0; σ := 1;
```

```
μx := 0
```

```
σ := 1
```

(8)

```
> Pb := evalf( ∫_{μx-σ}^{μx+σ} f(x) dx )
```

```
Pb := 0.6826894920
```

(9)

```
> Pa := evalf( ∫_{-∞}^{μx-σ} f(x) dx )
```

```
Pa := 0.1586552539
```

(10)

```
> Pc := evalf( ∫_{+∞}^{μx+σ} f(x) dx )
```

(11)

|

=

>

$$Pn := evalf\left(\int_{-\infty}^{+\infty} f(x) \, dx\right)$$

$Pc := -0.1586552539$

$Pn := 1.$

(11)

(12)